Lab instructions

Goal

The objective of this Lab is that you will be able to create a database and tables. Then, you will demonstrate how to work with data types in SQL.

❖ Task 1

Scenario

Mr. Mohamed Zaouali owns a small business named “MZ Mobiles” that sells mobile devices. He wants to create a database to store key information about mobile devices. This information includes:

- The device ID or number,
- The device name,
- And the price of the device.

This information is displayed in the following table:

Mobile Devices Table

Device ID	Device name	Price
1	iPhone XR 1	1500.50
2	Samsung SX	1200.50
3	Nokia 730	1050.00

***Note**: You are required to complete this lab inside MySQL .*

Download and install MySQL :: Download MySQL Community Server.

Instructions

1. Create a database called **mz_devices** and make sure you select the database you want to create the table inside.
2. Identify a suitable name for the table in which you want to store the data about mobile devices. In this case, you can call the table “**devices**”. Then, identify the data type for each column within the table.
3. Write a complete SQL statement to create the mobile devices table inside the **mz_devices** database.
4. Access the devices table that was created inside the mz_mobiles database
5. Check the structure of the devices table (Display all the devices table’s columns and data types.)

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❖ Task 2

Scenario

Mr. Mohamed Zaouali wants to create another basic table in the database to store data about the stock of the devices including device ID, quantity available in the stock and total available cost of the quantity. This basic table is shown in the table below, with each column showing device ID, the quantity in stock and the total price.

Stock table

Device ID	Quantity	Total price
1	5	5000.75
2	3	3500.50

Instructions

1. Identify an appropriate table name to create, given the information provided.
2. Identify columns that should be available in this table and define them with the appropriate data types.
3. Write the full SQL statement that creates the table and columns.
4. Access the table that was created
5. Display all the devices table's columns and data types.

❖ Task 3

Scenario

Mr. Mohamed Zaouali wants to create a new table to store key information about customers including:

- The customer username: It is always nine characters in length and it contains alphanumeric values.
- The Full Name: It must be more flexible in terms of the length as it can vary from one customer to another, with a maximum of 100 characters.
- The Email address: The maximum length of the email will be 255 characters.

Customer Table:

username	Full Name	Email
Custom001	John Johnson	J.Johnson@email.com
Custom002	Carl Schmidt	Carl.Schmidt@email.com
Custom003	Yara Suliman	Yara.Suliman@email.com

Instructions

1. Write a complete SQL statement to create the table "**customers**" inside the **mz_devices** database with relevant attributes and data types.
2. Display all the **customers** table's columns and data types.

❖ Task 4

Scenario

Mr. Mohamed Zaouali wants to create another basic table in the **mz_devices** database to store the customers' **feedback**. This table must include three columns:

- Feedback ID column,
- Feedback type column with a maximum of 100 characters,
- A comment column with a maximum of 500 characters.

The **feedback table** is illustrated below:

Feedback ID	Feedback type	Feedback comment
Feed0001	Good	It is a very good product but the delivery was not great.
Feed0002	Very good	No comment

You are required to complete the following tasks:

1. Declare the columns with the right data type for each.
2. Write the SQL statement that creates the feedback table.

❖ Task 5

Scenario

Mr. Mohamed Zaouali owns a small business that sells mobile devices called "MZ Mobiles" in Harrow, London. He wants to create a database to store key information about customers' addresses including customer ID, street, postcode and town name. The list of addresses for current customers of MZ Mobiles is listed in the address table below.

Address Table

Customer ID	Street	Postcode	Town
1	10 Carsten Rd.	HA3 0AE	Harrow
2	15 Denise Ave.	HA3 0AE	Harrow
3	13 Merkel Ave.	HA3 0AE	Harrow
4	12 Carsten Rd.	HA3 0AE	Harrow
5	15 Hellen way	HA3 0AE	Harrow
6	13 Carsten Rd	HA3 0AE	Harrow
7	11 Malika Rd.	NW9 0AX	Kingsbury

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Notice that the address table shows that most customers are living in Harrow area, which means MZ Mobiles customers are coming mainly from this city. In this case, you can define a *default value* for the city column as "Harrow" when you create the address table.

This would save Mr. Zaouali from having to enter "Harrow" repeatedly into the city field for each new customer record as it's filled automatically with the default value "Harrow". Of course, this assumes that no other value was entered in the table.

Instructions

Create an SQL statement with relevant attributes and constraints as follows:

1. Identify the column that requires default values.
2. Write a complete SQL statement to create the address table with relevant constraints.
3. Check the structure of the address table.

Task 6

Scenario

Mr. Mohamed Zaouali notices that most customers come from the same postcode i.e., "HA97DE".

Instructions

1. Write the SQL statement again to declare both the postcode and the town name with default values.
2. Drop the table **Address;**

Task 7

Scenario

Mr. Mohamed Zaouali wants to create a database to store key information about customers' orders in order to generate invoices for his customers including:

- Customer name: the string data type would be very suitable for the customer's name as we expect it to store data with alphabetic characters. In this case we can use VARCHAR as we expect to have customer names with different lengths of characters.
- Order date: the date data type would be very suitable for the order date as we expect it to store data with dates.
- Product quantity: the integer data type would be very suitable for the order quantity as we expect it to store whole numbers of data.

- Total price: the decimal data type would be very suitable for the product price as we expect it to store numeric data with fractions.

This data can be seen in the following invoice table:

Invoice Table

Customer name	Order date	Product quantity	Total price
Mark Dani	25/03/2022	5	750.50
Karl Masonry	22/03/2022	4	600.75
Jack Raymond	20/03/2022	3	978.00

Instructions

1. Write a complete SQL statement to create the **invoices** table inside the **mz_devices** database.
2. Display the invoices table created inside the mz_devices database.
3. Check the structure of the invoices table.

Task 8

Scenario

Mr. Mohamed Zaouali needs to have a new table to store the contact details of each customer including customer account number, customer phone number and customer email address.

1. You are required to choose a relevant data type for each of the columns.
2. Write a complete SQL statement to create the **contact** table inside the **mz_devices** database.